

Applied Bioinformatics Laboratory

Experiment – 3

Immunoinformatics Analysis of Antigenic Epitopes: B-Cell and T-Cell Epitope Prediction using IEDB

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AIM:

To computationally identify promising T-cell and B-cell epitopes from a selected antigen sequence using the IEDB Analysis Resource, with the objective of selecting immunogenic regions useful for vaccine design and immune response evaluation.

INTRODUCTION:

Immunoinformatics combines bioinformatics approaches with immunological principles to support rapid and cost-effective vaccine discovery. One of the most important steps in this field is epitope prediction, which involves locating antigenic regions capable of triggering immune responses. The Immune Epitope Database (IEDB) offers a comprehensive suite of tools to predict MHC-binding peptides (T-cell epitopes) and linear or conformational B-cell epitopes. These computational predictions significantly reduce experimental workload by prioritizing the most promising antigenic regions for laboratory validation.

METHODOLOGY:

- The amino acid sequence of the target protein was obtained in FASTA format from public repositories such as UniProt or NCBI Protein.

B-Cell Epitope Prediction:

- The B-cell epitope prediction module available on the IEDB platform was used.
- The antigen sequence was uploaded, and prediction methods such as Kolaskar & Tongaonkar Antigenicity or BepiPred-2.0 were selected to identify surface-exposed and antigenic regions.

T-Cell Epitope Prediction (MHC Class I and II):

- The protein sequence was submitted to the MHC Class I and MHC Class II binding prediction tools.
- Relevant HLA alleles (for example, HLA-A*02:01) were chosen based on population coverage and immunological relevance.
- The IEDB recommended or consensus prediction settings were applied to improve prediction reliability and accuracy.
- Predicted peptides were filtered using percentile rank values and IC₅₀ thresholds (IC₅₀ < 500 nM for MHC Class I), where lower values indicated stronger peptide–MHC binding affinity.

Conservation and Safety Analysis:

- Selected epitopes were evaluated for sequence conservation across multiple pathogen strains.
- Additional screening was performed to assess potential toxicity and allergenicity to ensure safety.

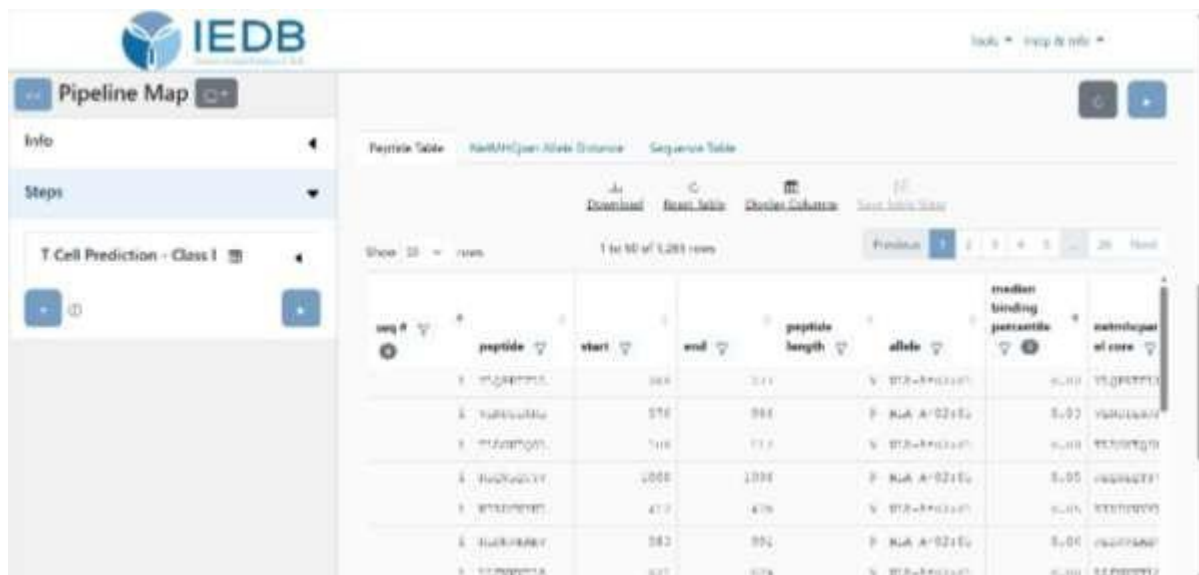
Visualization and Epitope Selection:

- High-scoring epitopes were visualized using IEDB graphical outputs.
- The most immunogenic candidates were shortlisted for further in-silico vaccine development studies.

QUESTION:

Immunoinformatics uses computational methods to predict antigenic regions for vaccine and immune response studies.

RESULTS:



seq #	peptide	start	end	peptide length	allele	median binding percentile	motif
1	YQPTPTT	588	613	26	HLA-A*02:01	0.10	YQPTPTT
2	YQPTPTT	576	598	23	HLA-A*02:01	0.02	YQPTPTT
3	YQPTPTT	518	543	26	HLA-A*02:01	0.10	YQPTPTT
4	YQPTPTT	1000	1026	27	HLA-A*02:01	0.05	YQPTPTT
5	YQPTPTT	472	498	27	HLA-A*02:01	0.10	YQPTPTT
6	YQPTPTT	582	604	23	HLA-A*02:01	0.00	YQPTPTT

Fig 1: IEDB MHC Class I T-Cell Epitope Prediction Output Showing Predicted Peptides, HLA Alleles, and Binding Percentile Ranks



Center position	Threshold	Predicted epitope
4	0.550	YQPTPTT

Fig 1.2: BepiPred Linear B-cell Epitope Prediction Results (IEDB Analysis Resource)

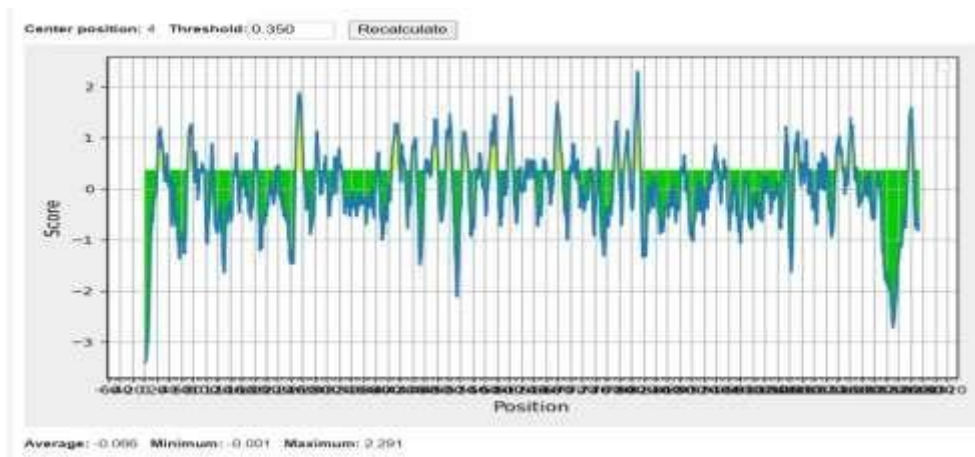


Fig 1.3: BepiPred Epitope Prediction Score Plot

Predicted peptides:

No.	Start	End	Peptide	Length
1	21	31	RTQLPPAYTNS	11
2	37	38	YY	2
3	71	81	SGTNGTKRFDN	11
4	84	85	LP	2
5	91	91	Y	1
6	95	95	T	1
7	97	98	KS	2
8	110	113	LDSK	4
9	150	152	KSW	3
10	181	186	GKQGNF	6
11	218	219	QG	2
12	221	221	S	1
13	249	261	LTFGDSSSGWTAG	13
14	282	287	NGTITD	6
15	296	298	LSE	3
16	312	313	IY	2
17	318	324	FRVQPTTE	7
18	382	385	VSPT	4
19	407	420	VRQIAPGQTGKIAD	14
20	423	428	YKLPDD	6
21	439	447	NNLDSKVGG	9
22	461	463	LKP	3
23	466	467	RD	2
24	469	469	S	1
25	473	483	YQAGSTPCNGV	11
26	495	506	YGFQPTNGVGYQ	12
27	523	532	TVCGPKKSTN	10
28	546	546	L	1

Fig 1.4: Predicted Peptides and Their Lengths

INTERPRETATION:

The immunoinformatics-based analysis successfully identified multiple high-affinity T-cell and B-cell epitopes within the target antigen. Epitopes showing low percentile ranks and strong MHC binding affinity indicate a high likelihood of immune recognition. The presence of conserved and non-toxic epitopes suggests their suitability as potential vaccine candidates. Overall, the IEDB-based epitope prediction approach provides a reliable and efficient strategy to prioritize immunogenic regions, thereby supporting rational vaccine design and reducing dependence on extensive experimental screening.